Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2001-I-Q02] | Part A (Q1-Q5)

Answer: 651

1. Neither adjoined value was already present, because otherwise its union would not change the mean. If the set has n elements, then $(nM+1)/(n+1) = M-13$ and $(nM+2001)/(n+1) = M+27$.

2. These become $M-1 = 13(n+1)$ and $2001-M = 27(n+1)$. Adding gives $n+1 = 50$, and $M = 1+13\cdot 50 = 651$.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2021-II-Q01] | Part A (Q1-Q5)

Answer: 550

1. Each palindrome is $101a + 10b$ for $1 \leq a \leq 9, 0 \leq b \leq 9$, with every pair occurring once.

2. The mean digits are 5 and $9/2$, so the mean number is $101 \cdot 5 + 10 \cdot 9/2 = 550$.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2024-II-Q02] | Part A (Q1-Q5)

Answer: 236

1. The length is even, and 9 appears two or three times. If it appears three times, the remaining entries sum to 3; the possible even lengths either have median 9 or use three 1s, tying the mode. Hence exactly two 9s occur and all other entries are distinct.

2. For four entries, the other entries sum to 12. If one exceeds 9, the middle entries are both 9, which fails. Otherwise write $a < b < 9$; the median $(b + 9)/2$ must be integral, so $b = 7, a = 5$.

3. For six entries the four other distinct positives sum to 12, giving only $\{1, 2, 3, 6\}$ or $\{1, 2, 4, 5\}$; both yield median $9/2$. Eight or more entries require six distinct positives summing to at least $21 > 12$. Thus the list is $5, 7, 9, 9$ and the squared sum is 236.

Source / 来源

Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2017-I-Q11] | Part C (Q11-Q15)

Answer: 360

1. Write the four digits below5 as L and the four above5 as G. The row containing5 must have one L and one G. The remaining two row medians must straddle5.

2. The other row types are either LLL and GGG, or LLG and LGG. Assigning the three row types and ordering the L5G row gives $3! \cdot 3! = 36$ patterns. The two cases supply another factor1 or9, respectively, for360 letter patterns.

3. Fill the L and G slots independently in $4!$ ways each. Thus $Q = 360(4!)^2 = 207360$, with remainder360.

Source / 来源